

Lecture 17

Orthonormal Basis and Orthogonal Matrix

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Today's Lecture: Outline

Main topic: Orthonormal basis

1. Orthonormal basis
2. Orthogonal matrix

Strang's book: mainly Sec 4.4; a bit of Sec 4.2

After the lecture, you should be able to

1. Prove the results on normal equations, with the aid of a lemma
2. tell the definition and properties of orthogonal matrix
3. compute the projection onto a subspace

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Roadmap of MATH2041

Segment 1 Solving linear systems (Lec 3-15)

Segment 2 [Lec 16-20]: Three relatively independent parts:

- Solving [least squares problem & orthogonality](#) (Lec 16, [17](#), 18)
- Determinant. [Important tool!] (Lec 19)
- [Linear transformation](#). [Advanced math perspective of [matrix](#)] (Lec 20,21)

Segment 3 Lec 21-27: Eigenvalues and related.

- [Eigenvalues](#). Lec 22-24
- [Singular values](#). Lec 25-26
- Quadratic forms. Lec 27.

Review

Recall: Orthogonality and LS Solution

Definition in Lec 16 (Orthogonality)

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if $\mathbf{u}^\top \mathbf{v} = 0$. Denote $\mathbf{u} \perp \mathbf{v}$

Thm 16.2 and Lemma 16.1 together:

Consider a least squares problem. The following statements are equivalent:

1. $\mathbf{y}$ minimizes $\| A\mathbf{x} - \mathbf{b} \|$
2. $\mathbf{b} - A\mathbf{y} \perp S$, where $S = \mathcal{C}(A)$.
3. $A^\top A\mathbf{y} = A^\top \mathbf{b}$

Part 0 Analyzing Normal Equation

17 mins

Solving Least Squares: Expression for Full-Rank A

Corollary (Characterization)

Suppose A has **linearly independent columns** (i.e., has full column rank).

Then solution of the least square problem $\min_{\mathbf{x} \in \mathbb{R}^{n \times 1}} \|A\mathbf{x} - \mathbf{b}\|$ is

$$\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$$

Remark: A has linearly independent columns $\iff \text{rank}(A) = n$

Proof: **Step 1:** Prove “invertible”.

Lemma 1: If A has full column rank, then $A^T A$ is invertible.

Step 2: Solution of $A^T A \mathbf{y} = A^T \mathbf{b}$ is $(A^T A)^{-1} A^T \mathbf{b}$

Proof of Lemma 1

Lemma 17.0: For any matrix A , $N(A^T A) = N(A)$

Proof: See Tutorial 8.

Lemma 1: If A has full column rank, then $A^T A$ is invertible.

A has full column rank $\rightarrow$ matrix (column space)

$\Leftrightarrow$ col's indep. $\Leftrightarrow \sum_{i=1}^n \alpha_i \vec{a}_i = 0 \Rightarrow \alpha_i = 0, \forall i$ $\rightarrow$ equation

$\Leftrightarrow A x = 0$ has exactly one solution $x = 0$ $\rightarrow$ linear space

$\Leftrightarrow N(A) = \{0\}$

$A^T A$ is invertible (square matrix) $\Leftrightarrow A^T A$ has full column rank

$\Leftrightarrow A^T A x = 0$ has exactly one solution $x = 0$

$\Leftrightarrow N(A^T A) = \{0\}$

Solving Least Squares: Solution

Proposition (Existence)

The linear system $A^T A \mathbf{y} = A^T \mathbf{b}$ has at least one solution.

Proof: $B \mathbf{x} \in \mathcal{R}, \forall \mathbf{x} \in \mathbb{R}^n \Leftrightarrow [C(B)] \subseteq \mathcal{R}$
 (any LC of columns of $B \in \mathcal{R}$)

Hint: Use the following fact:
" $B\mathbf{x}=\mathbf{z}$ " is solvable iff $\mathbf{z} \in C(B)$.

Want to prove: $A^T \mathbf{b} \in C(A^T A)$ for all $\mathbf{b}$,

which is equivalent to: $C(A^T) \subseteq C(A^T A) \iff \text{Row}(A) \subseteq \text{Row}((A^T A)^T). \quad (1)$

Derive from what we know.

From Lemma 17.0: $N(A) = N(A^T A)$.

orthogonal complement $\Rightarrow \text{Row}(A)^\perp = \text{Row}(A^T A)^\perp \Rightarrow \text{Row}(A) = \text{Row}(A^T A). \Rightarrow (1)$

since $N(A) = \text{Row}(A)^\perp$ (Lec 15)

Part I Orthogonal Set

20 mins

Orthonormal Set of Vectors

Definition 17.1 (Orthogonal Set)

Let $\{v_1, v_2, \dots, v_k\}$ be a set of **nonzero** vectors in $\mathbb{R}^n$. If $\langle v_i, v_j \rangle = 0$ for any $i \neq j$, then this set is called an **orthogonal set**.

Examples:

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \quad \left\{ \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} -4 \\ 3 \end{pmatrix} \right\}$$

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

norm 2

Definition 17.2 (Orthonormal Set)

An orthonormal set $\{v_1, v_2, \dots, v_k\}$ is an orthogonal set of unit-norm vectors, i.e., $\|v_i\| = 1$ for all $i = 1, \dots, n$.

Examples:

$$\left\{ \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$\left\{ \begin{bmatrix} 3 \\ 5 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ -5 \\ 3 \\ 5 \end{bmatrix} \right\}$$

norm 1 norm 1

$$\left\{ \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

From an Orthogonal Set to an Orthonormal Set

Question

Can you create an **orthonormal set** based on an **orthogonal set**?

$$\{u_1, u_2, \dots, u_n\}$$



$$\left\{ \frac{u_1}{\|u_1\|}, \dots, \frac{u_n}{\|u_n\|} \right\}$$

Orthogonal set



Orthonormal set

Note: $\|u_j\| \neq 0$ by Definition of orthogonal set

Examples

Example

Consider $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ in $\mathbb{R}^3$

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 4 \\ -5 \\ 1 \end{bmatrix}$$

then $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set, the orthonormal set is

$$\left\{ \mathbf{v}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \frac{1}{\sqrt{14}} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \frac{1}{\sqrt{42}} \begin{bmatrix} 4 \\ -5 \\ 1 \end{bmatrix} \right\}$$

What is Special about an Orthogonal Set?

What words can be used to describe a set of vectors?

General learning tip:
when learn new things,
relate to your knowledge.

a set of vectors cannot be symmetric.

a set of vectors cannot be invertible.

a set of vectors cannot be upper triangular.

a set of vectors can be

~~matrix~~ independent

a set of vectors can be a

basis

a set of vectors can be a

spanning set

What is Special about an Orthogonal Set?

What words can be used to describe a set of vectors?

Orthogonal v.s. independent. (next page)

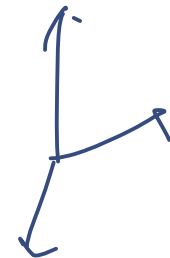
Orthogonal v.s. basis. (next part)

General learning tip:
when learn new things,
relate to your knowledge.

Question If $\{v_1, v_2, \dots, v_k\}$ is an orthogonal set, is it an independent set?

Yes.

Let students vote.



Proof of Proposition 17.1

Proposition 17.1

Let $S = \{v_1, \dots, v_k\}$ be an orthogonal set. Then $v_1, \dots, v_k$ are linearly independent.

In words: Vectors in an orthogonal set are linearly independent

Proof for $k=2$, i.e.: Let $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$ be non-zero vectors such that $\vec{v}_1^\top \vec{v}_2 = 0$. Then $\{\vec{v}_1, \vec{v}_2\}$ is linearly independent.

Assume $\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 = \vec{0}$.

relation?

Multiply by $\vec{v}_1^\top$: $\alpha_1 \vec{v}_1^\top \vec{v}_1 + \alpha_2 \vec{v}_1^\top \vec{v}_2 = 0 \Rightarrow \alpha_1 \|\vec{v}_1\|^2 = 0$

Since $\vec{v}_1^\top \vec{v}_2 = 0$ and $\|\vec{v}_1\|^2 \neq 0$, we get $\alpha_1 = 0$.

or $>$ (Since $v_1 \neq 0$ in orthogonal set).

Similarly, multiply by $\vec{v}_2^\top$: $\alpha_2 \|\vec{v}_2\|^2 = 0 \Rightarrow \alpha_2 = 0$

Thus $\alpha_1 = \alpha_2 = 0$, so the set is **linearly independent**.

Part II Orthonormal Basis and Orthogonal Matrix

20 mins

Orthonormal Basis (标准正交基)

What words can be used to describe a set of vectors?

Orthogonal v.s. **independent**. (previous page)

Orthogonal v.s. basis. (next page)

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

An orthogonal set may not be a basis. [Example?]

We combine orthonormal set & basis: create a new concept.

Orthonormal Basis (标准正交基)

Definition 17.3

A set of vectors $S = \{v_1, \dots, v_n\}$ is called an **orthonormal basis** of a linear space V if

- (1) S is an orthonormal set
- (2) Vectors in S form a basis of V .

Example

(1). Standard basis

(2). $\left\{ \mathbf{v}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \frac{1}{\sqrt{14}} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \frac{1}{\sqrt{42}} \begin{bmatrix} 4 \\ -5 \\ 1 \end{bmatrix} \right\}$ for $\mathbb{R}^3$

↓
orthonormal
(seen before)

basis (3 vec in $\mathbb{R}^3$,
independent)

Orthogonal Matrix

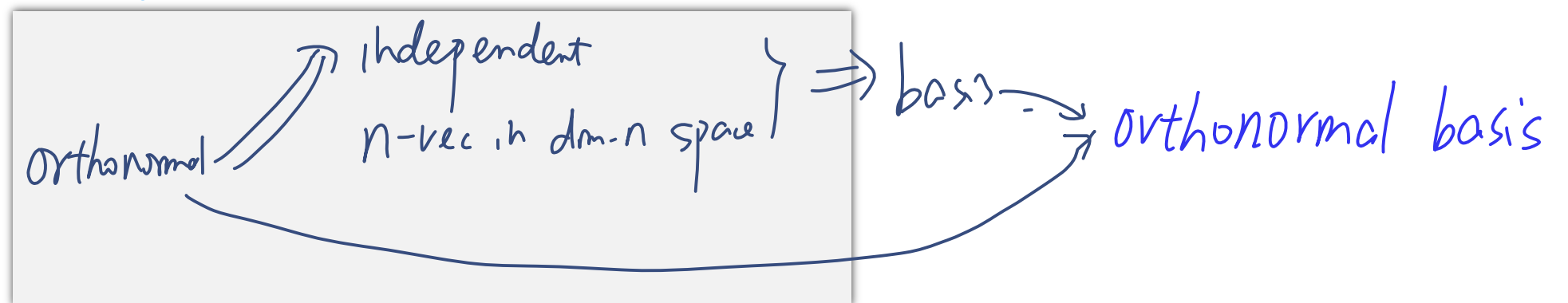
Definition 17.4 (Orthogonal Matrix)

An orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ is a real square matrix whose columns form an **orthonormal basis** in $\mathbb{R}^n$

Examples:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & \frac{4}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & \frac{-5}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & \frac{1}{\sqrt{42}} \end{bmatrix}$$

Claim: If $\dim(V) = n$, then an orthonormal set $S = \{v_1, \dots, v_n\}$ is an orthonormal basis.
[based on what result of past lectures?]



Corollary: If the columns of $Q \in \mathbb{R}^{n \times n}$ form an **orthonormal set** of $\mathbb{R}^n$, then it is an orthogonal matrix.

Proof of Proposition 17.4

Suggested break

Proposition 17.2 (Orthogonal Matrix)

A matrix $Q \in \mathbb{R}^{n \times n}$ is an orthogonal matrix if and only if $Q^T Q = I_n$.

Hint: Find a proper form of matrix multiplication.

Proof: Let $Q = [q_1 \ \cdots \ q_n] = \begin{bmatrix} q_{(1)}^T \\ \vdots \\ q_{(n)}^T \end{bmatrix}$, where q_i are the columns of Q , and $q_{(i)}^T$ are the rows of Q .

$$Q^T Q = \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} [q_1 \ \cdots \ q_n] = \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \cdots & q_1^T q_n \\ \vdots & \vdots & \ddots & \vdots \\ q_n^T q_1 & q_n^T q_2 & \cdots & q_n^T q_n \end{bmatrix} = I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$\Leftrightarrow \begin{cases} q_i^T q_j = 0, & \forall i \neq j \quad (\text{orthogonal columns}) \\ q_i^T q_i = 1, & \forall i \quad (\text{unit-norm columns}) \end{cases}$$

$$\Leftrightarrow \{q_i\} \text{ is an orthonormal set} \iff Q \text{ is an orthogonal matrix.}$$

Orthogonal Matrix

Claim If $Q \in \mathbb{R}^{n \times n}$ satisfies $Q^T Q = I_n$, then $Q Q^T = I_n$.

Proof: Step 1: Q is invertible since columns of Q are

orthogonal
 $\Rightarrow$ independent.

Step 2: $Q Q^T = \underbrace{Q Q^T}_{I} \underbrace{(Q Q^{-1})}_{I} = \underbrace{Q Q^{-1}}_{I} = I_n.$

Corollary: $Q^{-1} = Q^T$

Another way: $Q^T = Q^T Q (Q^{-1}) = I \cdot Q^{-1}$

Example For any θ

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$Q^{-1} = Q^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

Properties of Orthogonal Matrix

These will be very useful
for eigenvalues after Lec 21.

Property 17.1 (Orthogonal Matrix)

If $Q \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, then

(a) The column vectors of Q form an orthonormal basis.

(b) $Q^T Q = I_n$.

(c) $Q^{-1} = Q^T$.

(d) $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$. [inner product preserving]

(e) $\|Q\mathbf{x}\| = \|\mathbf{x}\|$. [norm preserving]

Proof of (d): $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = (Q\mathbf{x})^T Q\mathbf{y} = \mathbf{x}^T \underbrace{Q^T Q}_{I} \mathbf{y} = \mathbf{x}^T \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$

Proof of (e): $\|Q\mathbf{x}\|^2 = \langle Q\mathbf{x}, Q\mathbf{x} \rangle \stackrel{(d)}{=} \langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$
or directly. $\|Q\mathbf{x}\|^2 = \mathbf{x}^T Q^T Q \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2$

Part III Representation and Projection

20 mins

Recall: Representing Element using Basis

Result of Lec 15: Full column rank $\Rightarrow$ at most one solution

Proposition 15.1: If $\text{rank}(A) = m$, i.e., full row rank, then $Ax = b$ has **at least one solution**.

Proof from vector perspective:

vectors u_i 's are independent $\rightarrow$ at most one way to represent any v by u_i 's.

Corollary 15.1 (Unique Representation via Basis)

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be a basis of a linear space V .

Any element $v \in V$ can be **uniquely** represented as a linear combination of $u_1, \dots, u_n$

[in Lec 15]

slightly diff from the version in Lec 15.
(you can compare)

Example:

$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Recall: Representing Element using Basis

Corollary 15.1 (Unique Representation via Basis)

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be a basis of a linear space V .
Any element $v \in V$ can be **uniquely** represented as a linear combination of $u_1, \dots, u_n$ [discussed in Lec 15]

Representation $v = \alpha_1 u_1 + \dots + \alpha_n u_n$.

We know such $\alpha_1, \dots, \alpha_n$ exist. But....how to compute?

$$\text{e.g. } \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

$$\vec{v} = U \cdot \vec{\alpha}$$

When $u_i \in \mathbb{R}^n$, finding $\alpha_1, \dots, \alpha_n$ is just

solving linear system.

$$Ax = b$$

Next, we show, if $\mathcal{U}$ is an **orthonormal** basis,
then finding $\alpha_1, \dots, \alpha_n$ is much easier.

Representing Element using Basis

Theorem 17.1 (Representation via Orthonormal Basis)

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be an orthonormal basis of $\mathbb{R}^n$.

Any vector v can be represented as a linear combination

$$v = \sum_{i=1}^n u_i u_i^\top v = \sum_{i=1}^n \langle v, u_i \rangle u_i$$

This means, $v = \alpha_1 u_1 + \dots + \alpha_n u_n$, where $\alpha_i = u_i^\top v$.

Proof Method 1 (vector):

From Coro. 15.1, there exist $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ s.t. $v = \alpha_1 u_1 + \dots + \alpha_n u_n$

Now compute α_i 's: For any $i \in \{1, \dots, n\}$, compute:

$$\begin{aligned} u_i^\top v &= u_i^\top (\alpha_1 u_1 + \dots + \alpha_n u_n) \\ &= \alpha_1 u_i^\top u_1 + \dots + \alpha_i u_i^\top u_i + \dots + \alpha_n u_i^\top u_n = \alpha_i \cdot 1 = \alpha_i. \end{aligned}$$

Here we use orthonormality:

- $u_i^\top u_j = 0$ for $i \neq j$ (orthogonal),
- $u_i^\top u_i = \|u_i\|^2 = 1$ (unit norm).

Representing Element using Basis

Theorem 17.1 (Representation via Orthonormal Basis)

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be an **orthonormal basis** of $\mathbb{R}^n$.

Any vector v can be represented as a linear combination

$$v = \sum_{i=1}^n u_i u_i^T v = \sum_{i=1}^n \langle v, u_i \rangle u_i$$

This means, $v = \alpha_1 u_1 + \dots + \alpha_n u_n$ (*), where $\alpha_i = u_i^T v$.

Proof Method 2 (matrix outer product; simplest):

Let $U = [u_1 \ \dots \ u_n]$ be a matrix whose columns form an **orthonormal basis**, so $UU^T = I$.

$$I = UU^T = \underbrace{[u_1 \ \dots \ u_n]}_{\text{block } 1 \times n} \underbrace{\begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}}_{\text{block } n \times 1} = \underbrace{u_1 u_1^T + \dots + u_n u_n^T}_{\text{block } 1 \times 1} = \sum_{i=1}^n u_i u_i^T$$

$$v = I \cdot v = (UU^T)v = \sum_{i=1}^n \underbrace{u_i}_{n \times 1} \underbrace{\left[\underbrace{u_i^T}_{1 \times n} v \right]}_{\substack{1 \times 1 \\ \text{scalar}}} = \sum_{i=1}^n \underbrace{(u_i^T v)}_{\text{scalar}} u_i.$$

Example of Representing using Orthonormal Basis

Application of $v = \sum_{i=1}^n \langle v, u_i \rangle u_i$.

Example

$$\left\{ \mathbf{v}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \frac{1}{\sqrt{14}} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \frac{1}{\sqrt{42}} \begin{bmatrix} 4 \\ -5 \\ 1 \end{bmatrix} \right\}$$

is an orthonormal basis of $\mathbb{R}^3$

For any $\mathbf{x} = [x, y, z]^T \in \mathbb{R}^3$, one has

$$\begin{aligned} \mathbf{x} &= \langle \mathbf{x}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{x}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{x}, \mathbf{v}_3 \rangle \mathbf{v}_3 \\ &= \frac{x + y + z}{\sqrt{3}} \mathbf{v}_1 + \frac{2x + y - 3z}{\sqrt{14}} \mathbf{v}_2 + \frac{4x - 5y + z}{\sqrt{42}} \mathbf{v}_3 \end{aligned}$$

Projection

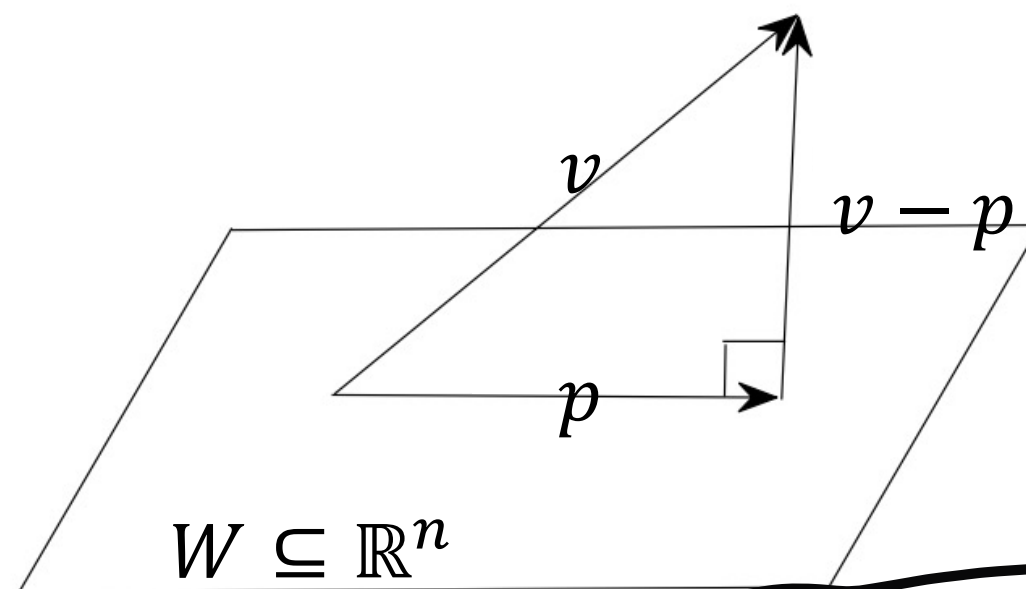
Definition 16.1 [Last lecture]

Suppose S is a subspace of $\mathbb{R}^m$.

Suppose $\mathbf{p} \in S$ and $\mathbf{b} - \mathbf{p} \perp S$, then we say $\mathbf{p}$ is the projection of $\mathbf{b}$ onto S .

Proposition 17.2 (projection exists and unique)

For any subspace $W \subset \mathbb{R}^n$, any vector $\mathbf{v} \in \mathbb{R}^n$, there is a **unique** vector $\mathbf{p} \in W$ such that $(\mathbf{v} - \mathbf{p}) \perp W$.



Skip the proof.

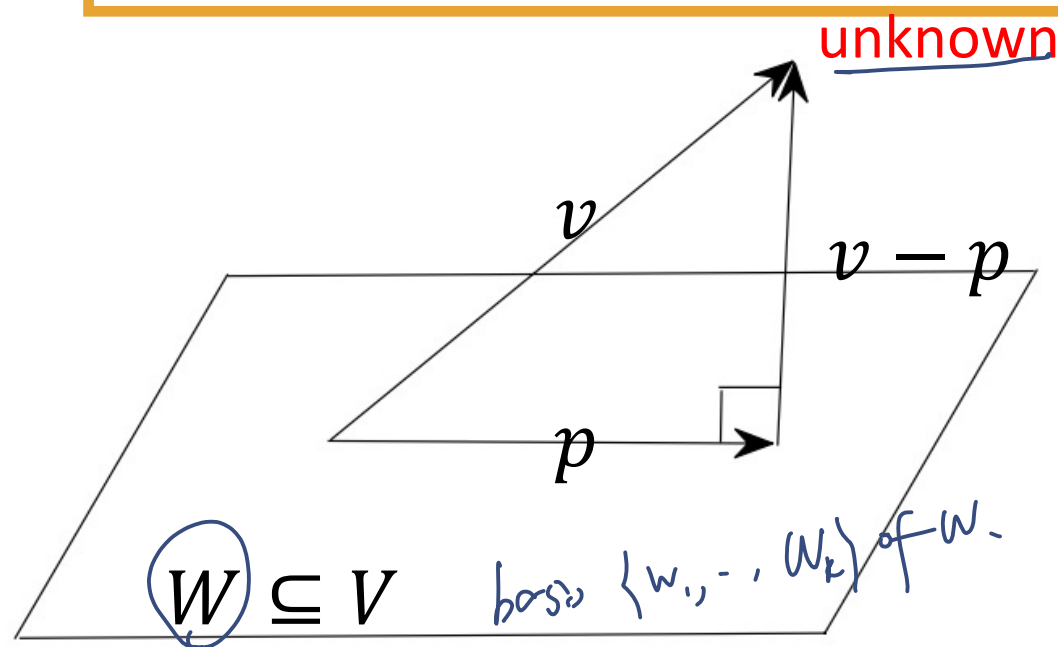
Next: Compute P .

Representing Projection via Orthonormal Basis

Theorem 17.2 (Compute Projection when Ortho-Basis Exists)

Let W be a subspace of $\mathbb{R}^n$ and suppose $S = \{w_1, \dots, w_k\}$ is an **orthonormal basis** of W . The projection of $\mathbf{v}$ onto W can be represented as

$$\mathbf{p} = \sum_{i=1}^k \langle w_i, \mathbf{v} \rangle w_i. (*)$$



Remark:

By Thm 17.1, we have $\mathbf{p} = \sum_{i=1}^k \langle w_i, \mathbf{p} \rangle w_i (**)$
 (*) and (**) differ: $\mathbf{v}$ v.s. $\mathbf{p}$.

What we want: From **known vector** $\mathbf{v}$,
 compute **unknown vector** $\mathbf{p}$. (*) achieves this goal.

In (**), RHS involves unknown $\mathbf{p}$,
cannot be used to compute projection $\mathbf{p}$.

Skip the proof.

Orthonormal Basis is Useful

Because

1. Representing any vector by directly computing the “coordinates” via computing inner products

—Useful in, e.g.,

computing LS-solution (next part)

computing orthonormal basis (next part)

2. Useful in eigen-theory (lecture 21)

But...does it exist, and how can we obtain an orthonormal basis?

日期:

Summary (By Student)

One sentence summary.

Detailed Summary:

Summary Today (of Instructor)

One sentence summary:

We have studied Orthonormal basis and Gram-Schmidt Process.

Detailed summary:

1) Orthonormal basis.

- Basis consisting of unit vectors that are orthogonal to each other.
- Theorem 17.1**: Representation of vector by ortho-basis of **whole space**.
- Theorem 17.2**: Representation of projection by ortho-space of a **subspace**.

2) Orthogonal matrix.

- Square matrix whose columns form a orthonormal basis of $\mathbb{R}^n$.
- Properties**: $Q^T Q = I$, keep the norm $\|Qx\| = \|x\|$;

3) Representation of vectors by orthonormal basis. $\vec{v} = \sum_{i=1}^n (\vec{v}^T \vec{u}_i) \vec{u}_i$
— Projection: $\vec{p} = \sum_{i=1}^k (\vec{v}^T \vec{w}_i) \vec{w}_i$, as projection of $\vec{v}$ onto W .

Summary Today (of Instructor)

One sentence summary:

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Detailed summary:

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